IND3001 Engineering Economics

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What is Economics?

- **Economics** is the study of how individuals, firms, and societies choose to use **scarce resources** to satisfy **unlimited wants**.
- The central problem is **scarcity. (time, money, material, labour)**
- Three basic questions: *what* to produce, *how* to produce it, and *for whom* to produce it.

The problem of scarcity

- **Scarcity** means that resources are limited relative to the wants people have for them.
- **Engineering examples of scarcity:**
- A municipality has a fixed budget and must choose between repairing a bridge, building a new road, or upgrading water pipes.
- A design team has 6 months and 10 engineers; they can't develop every feature the customer wants.
- A factory has one CNC machine that can run only 16 hours a day.

Choice

- Because resources are scarce, **every decision is a choice among alternatives**.
- Choosing one alternative means giving up others. This is the starting point of engineering economics: the engineer's job is to identify the alternatives and choose the best one.

Rational Choice and Thinking at the Margin

- Economists assume decision-makers are **rational**: they choose the option whose benefits exceed costs by the most. Most real decisions are **marginal**, meaning "a little more or a little less," not "all or nothing."

PRODUCTION

The transformation of inputs into outputs by firms in order to earn profit

Labour-wages
Land-Rent
Capital-Profit
+
Entrepreneurship

DISTRIBUTION

CONSUMPTION

The act of using goods and services to satisfy wants

Factors of Production /Land

- **Land** includes everything provided by nature: land surface, minerals, oil and gas, water, forests, fish stocks, sunlight, wind, and climate.
- Key characteristics:
- **Supply is largely fixed.** We cannot create more land, though we can use it more intensively.
- **Renewable vs non-renewable.** Forests, water, solar, and wind energy can be renewed; oil, coal, and metal ores are depleted when used.
- **Location matters.** The value of land depends heavily on where it is (a factory site near a port versus far inland).
- **Quality varies.** Fertile versus poor soil, high-grade versus low-grade ore.
- **Income: rent.** The payment for using land is rent, whether it is literally rented or owned

Factors of Production /Labour

- Labour: Human physical and mental effort, workforce
- Key characteristics:
- **Quantity** depends on population, working age, participation rate, and working hours.
- **Quality** depends on education, training, experience, and health. This is called **human capital**.
- **Labor is mobile but imperfectly so.** Workers can change jobs or cities, but it takes time and retraining.
- **Division of labor and specialization** increase productivity: when each worker specializes in one task, total output rises (the classic pin factory example from Adam Smith).
- **Income: wages and salaries.**

Factors of Production /Capital

- **Capital** refers to **human- goods used to produce other goods and services**.
- Types of capital:
- **Fixed capital:** long-lasting assets used repeatedly (buildings, machines, vehicles, cranes, production lines).
- **Circulating capital:** inputs used up in a single production cycle (raw materials, spare parts, work-in-progress inventory).
- **Infrastructure (social capital):** roads, ports, power grids, communication networks, usually public.

Factors of Production /Capital

- Key characteristics:
- **Capital requires sacrificing present consumption.** To build a machine, society must save instead of consuming today. This is why capital earns **interest**: it rewards waiting.
- **Capital depreciates.** Machines wear out and become obsolete. Depreciation is a major topic in engineering economics because it affects taxes and replacement decisions.
- **Capital increases productivity.** A worker with a forklift moves far more material than one without.
- **Income: interest.**
- !!!! Money is not capital in this sense. Money becomes capital only when it is used to buy productive assets !!!!

Factors of Production / Entrepreneurship

- **Entrepreneurship** is the ability and willingness to combine land, labor, and capital to produce goods and services, while taking risk.
- The entrepreneur's functions:
- **Organizing** the other factors of production.
- **Making key decisions**: what to produce, how, where, and how much.
- **Innovating**: introducing new products, processes, or markets.
- **Bearing risk**: if the business fails, the entrepreneur loses; the other factors have already been paid.
- **Income: profit.** Profit is a residual: what remains after rent, wages, and interest are paid. It can be negative (a loss), which is what makes entrepreneurship risky.

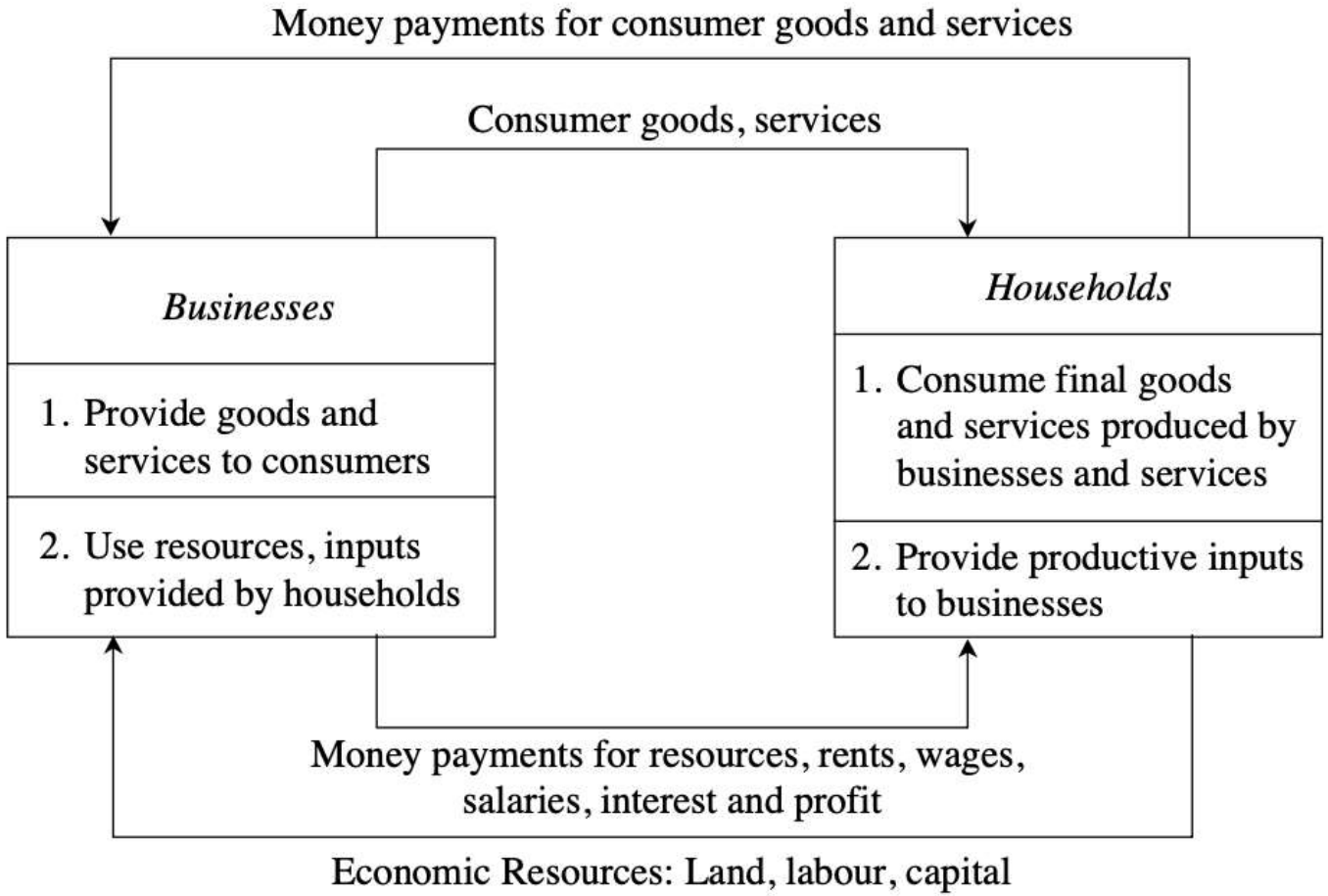
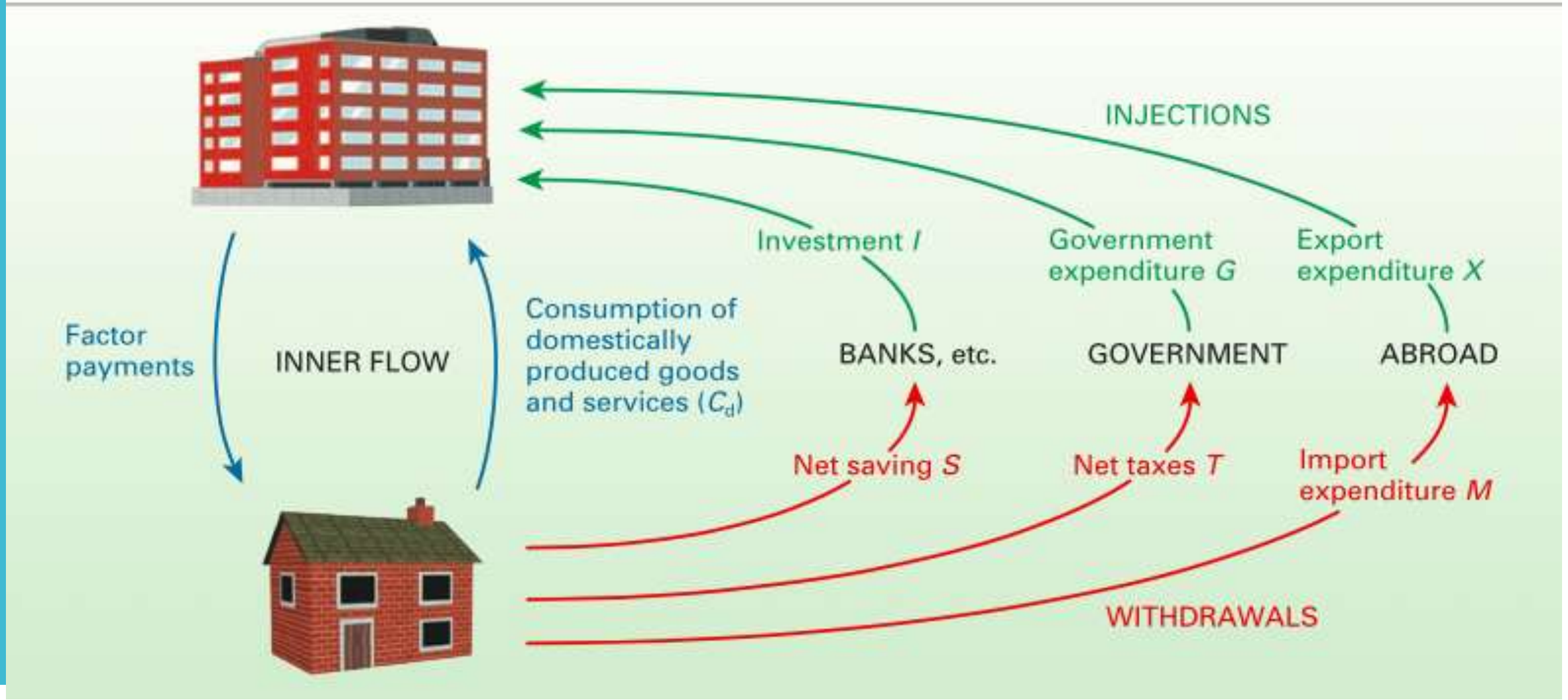


Fig. 1.1 Flow of goods, services, resources and money payments in a simple economy.

Figure 14.1 The circular flow of income



Law of Supply and Demand

- **Demand** is the quantity of a good that buyers are **willing and able** to purchase at each possible price, during a given period, holding other factors constant. Both conditions matter: wanting a car without being able to pay for it is not demand.
- Other things being equal, when the price of a good rises, the quantity demanded falls; when the price falls, the quantity demanded rises.
- Price and quantity demanded move in **opposite directions** (an inverse relationship). Three reasons explain why:
- **Substitution effect:** when a good becomes more expensive, buyers switch to substitutes (if aluminum rises in price, some designers switch to composites or steel).
- **Income effect:** a higher price reduces buyers' purchasing power, so they can afford less.
- **Diminishing marginal utility:** each additional unit gives less satisfaction, so buyers will only purchase more if the price is lower.

Determinants of Demand (What Shifts the Demand Curve)

- Price changes cause a **movement along** the demand curve. Changes in anything else cause the **entire curve to shift**. The main determinants are:
- **Income.** For **normal goods**, higher income increases demand. For **inferior goods**, higher income decreases demand.
- **Price and quantity of substitue goods.** Substitute goods give same utility when they consumed.
- **Price and quantity of complemantary goods.** They are purchased together.
- **Tastes and preferences.** Growing environmental awareness increases demand for solar panels.
- **Expectations.** If buyers expect prices to rise next month, they buy more now.
- **Number of buyers.** Population growth or a new market opening increases demand.
- **Government policy.** Subsidies or tax credits for buyers (e.g., solar incentives) increase demand.

Law of Supply and Demand

- **Supply** is the quantity of a good that producers are **willing and able** to offer for sale at each possible price, during a given period, holding other factors constant.
- Other things being equal, when the price of a good rises, the quantity supplied rises; when the price falls, the quantity supplied falls.
- Price and quantity supplied move in the **same direction** (a direct relationship). The reasons:
- **Profit incentive:** a higher price makes production more profitable, so firms produce more.
- **Increasing marginal cost:** producing extra units costs more (overtime, less efficient machines, costlier inputs), so firms need a higher price to cover those costs. This links back to the law of diminishing returns from the factors of production section.
- **New entrants:** higher prices attract new firms into the market.

Determinants of Supply (What Shifts the Supply Curve)

- **Input prices.** Higher costs for raw materials, labor, or energy reduce supply (curve shifts left).
- **Technology.** Better production technology lowers costs and increases supply (shift right).
- **Prices of other goods the firm could produce.** If a factory can make either panels or batteries and battery prices rise, it may shift capacity to batteries, reducing panel supply.
- **Expectations.** If producers expect higher prices later, they may hold back supply now.
- **Number of producers.** More firms mean more supply.
- **Government policy.** Taxes and regulations raise costs and reduce supply; subsidies to producers increase it.
- **Natural conditions and disruptions.** Earthquakes, floods, pandemics, and shipping blockages reduce supply.

Demand and Supply Curve

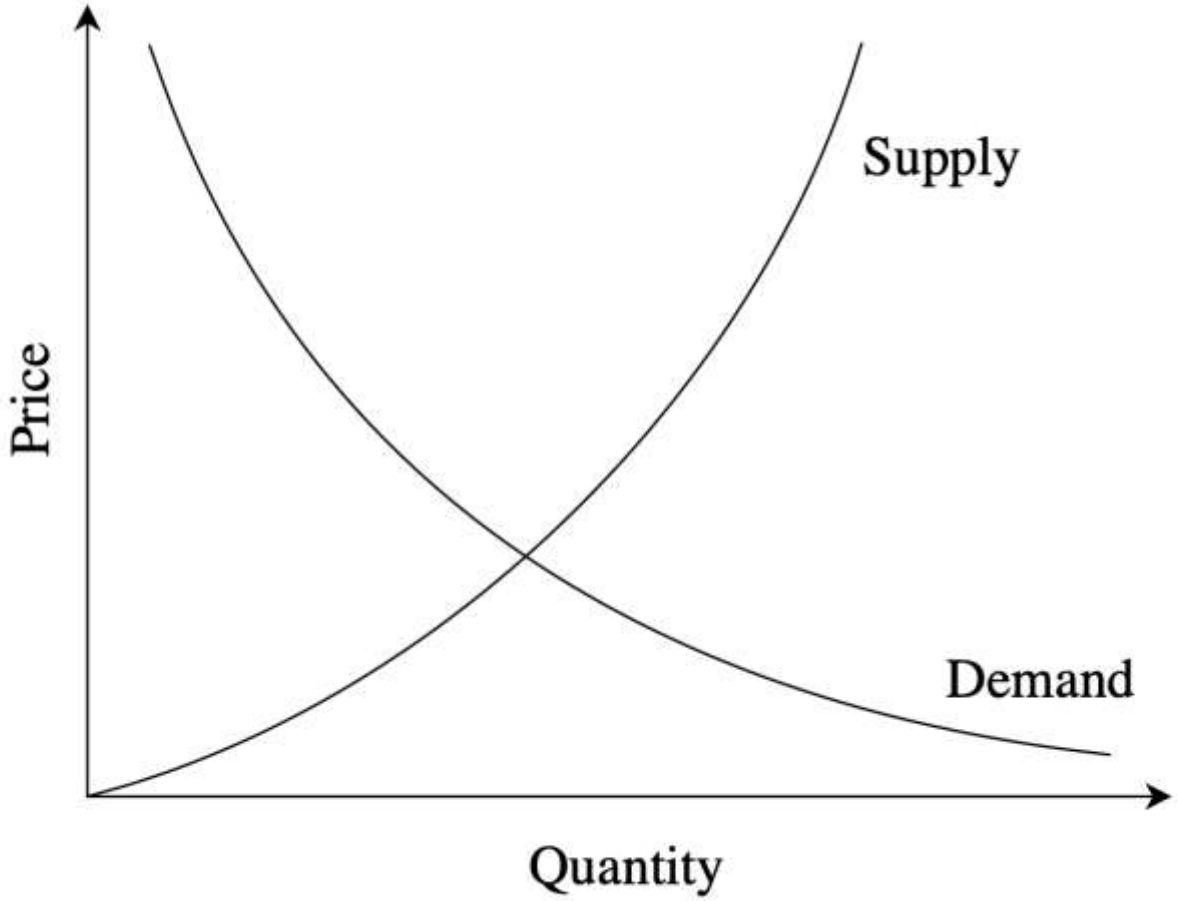


Fig. 1.2 Demand and supply curve.

Market Equilibrium

- **Equilibrium** is the price at which **quantity demanded equals quantity supplied**. At this price, the market "clears": everyone who wants to buy at that price can, and everyone who wants to sell can.

Price ($)	Qd	Qs	Market condition
20	800	50	Shortage of 750
30	700	200	Shortage of 500
40	600	350	Shortage of 250
50	**500**	**500**	**Equilibrium**
60	400	650	Surplus of 250
70	300	800	Surplus of 500
80	200	950	Surplus of 750

Solar Panel Market: Equilibrium at $50 and 500 Panels

Price ($/panel) by Quantity (panels/week)





Capitalism

Capitalism is an economic system in which the **means of production** (land, factories, machines, tools) are **privately owned**, production is organized for **sale in markets**, and the main goal of production is **profit**.

Key features

- **Private ownership of the means of production.** Factories and equipment belong to individuals or companies, not the state or the community.
- **Wage labor.** Most people do not own means of production, so they earn a living by selling their ability to work in exchange for wages.
- **Market exchange.** Goods and services are produced as commodities to be sold; prices are formed through supply and demand.
- **Profit motive.** Firms produce not primarily to satisfy needs directly but to earn profit.
- **Competition.** Firms compete for customers, markets, and resources, which pressures them to cut costs and innovate.
- **Capital accumulation.** Profits are reinvested to expand production, so the system tends to grow continuously.

Historical Emergence

- Capitalism developed gradually in Western Europe between the 16th and 19th centuries, replacing **feudalism**, a system in which peasants were tied to the land and handed over part of their produce to lords.
- The **Industrial Revolution** (from roughly the late 18th century in Britain) turned capitalism into the dominant system, with factories, machinery, and a large industrial working class.

Commodity

- Marx begins *Capital* with the **commodity**: a product made for exchange. Every commodity has two aspects:
- **Use value:** its usefulness, its ability to satisfy a need
- **Exchange value:** the proportion in which it trades for other commodities

- Commodities are all **products of human labor**.

Socially Necessary Labor Time

- The **value** of a commodity is determined by the **socially necessary labor time** required to produce it: the time needed under normal conditions of production, with average skill and intensity, using the typical technology of the time.
- **Two Circuits: C–M–C and M–C–M′**
- Marx contrasts two forms of circulation:
- **C–M–C (Commodity–Money–Commodity):** a shoemaker sells shoes for money, then uses the money to buy bread. The goal is a use value; money is just a go-between. This is "selling in order to buy."
- **M–C–M′ (Money–Commodity–More money):** a capitalist invests money, buys commodities (materials, machines, labor), and sells the product for **more money** than was invested. The goal is not a use value but **an increase in value**. This is "buying in order to sell."

Necessary Labor and Surplus Labor

- Divide the working day into two parts:
- **Necessary labor time:** the hours during which the worker produces value equal to their wage.
- **Surplus labor time:** the remaining hours, during which the worker produces value that goes to the capitalist without an equivalent payment. This is **surplus value**.
- **Example:** A worker in a machine shop works a 10-hour day. Suppose the value produced in 4 hours equals the daily wage. The remaining 6 hours produce surplus value.

Constant & Variable Capital

- Marx divides the capital a firm invests into two parts:
- **Constant capital (c):** money spent on means of production: machinery, buildings, raw materials, energy. Their value is **transferred** to the product (machines gradually, through wear; materials entirely) but **not increased**. Hence "constant."
- **Variable capital (v):** money spent on wages (labor power). This part **creates new value**, including surplus value, so its size "varies" in the production process.

Absolute and Relative Surplus Value

- Capitalists can increase surplus value in two ways:
- **Absolute surplus value:** lengthening the working day or intensifying work. If the day in our example rises from 10 to 12 hours with the same wage, surplus labor goes from 6 to 8 hours. This faces physical limits and resistance from workers and legislation. Marx devoted long chapters to the 19th-century British struggles over the length of the working day and the Factory Acts.
- **Relative surplus value:** reducing the **necessary** labor time by raising productivity. If technology makes workers' consumer goods cheaper, the value of labor power falls, so the worker covers their wage in, say, 3 hours instead of 4, and surplus labor grows to 7 hours without lengthening the day.

- Relative surplus value is where **technology and engineering** enter Marx's analysis!!

Capital Accumulation

- **Capital accumulation** means converting surplus value back into capital: using profits to buy more means of production and hire more labor, so production expands.
- Marx distinguishes:
- **Simple reproduction:** the capitalist consumes all surplus value personally. Production repeats on the same scale each period.
- **Expanded reproduction (accumulation):** part of the surplus value is reinvested, so each period production takes place on a larger scale.

Why Must Capitalists Accumulate?

- A central point in Marx's argument is that accumulation is **not a personal choice** based on greed. It is **forced by competition**. A firm that does not reinvest in newer, more productive machinery will be undersold by rivals and eventually driven out of the market. Competition acts as an external, coercive law on every individual capitalist.

Consequences of Accumulation

- Marx drew several long-run tendencies from this analysis:
- **(a) Concentration and centralization of capital.** *Concentration* is the growth of individual capitals through reinvestment. *Centralization* is the merging of capitals, where large firms absorb smaller ones through competition, takeovers, and the credit system. Marx predicted industries would come to be dominated by fewer, larger firms.
- **(b) Rising organic composition of capital.** Competition drives firms to mechanize, so spending on machinery (c) grows faster than spending on labor (v). The ratio c/v tends to rise.
- **(c) The industrial reserve army of labor.** Mechanization replaces workers, creating a pool of unemployed and underemployed people. Marx argued this "reserve army" serves capital by holding down wages and providing workers who can be hired quickly when the economy expands. So accumulation can grow employment in absolute terms while still regularly producing unemployment.
- **(d) The tendency of the rate of profit to fall (TRPF).** This follows from (b). Since, in Marx's theory, only living labor creates surplus value, if the share of capital spent on labor shrinks, the profit rate tends to fall, assuming the rate of surplus value is unchanged.
- **(e) Crises.** Because each firm expands production independently, without coordination, and because workers' purchasing power is limited relative to what they produce, Marx argued capitalism suffers recurring **crises of overproduction**: more goods are produced than can be sold at a profit. Crises destroy and devalue capital (bankruptcies, idle factories, falling asset prices), which can restore profitability and start a new cycle of accumulation.
- **(f) Growing polarization.** Marx expected wealth to accumulate at one pole (owners of capital) while insecurity and hardship would accumulate at the other (the working class), leading ultimately to political conflict and, in his view, a transition beyond capitalism.

Vivek Chibber, *Confronting Capitalism: How the World Works and How To Change It*

- *«Capitalists don't cause harm to their employees out of malice or greed. Their motivation comes from the **brute reality of market competition.** If a capitalist doesn't produce at the lowest price, he knows that he will lose customers, and if that continues, his firm will start bleeding money. So he has to keep his selling price as low as possible. But if he's going to lower his selling price, he also has to lower his* costs, *or he won't make any money. Hence, he tries to pay out as little as possible for his inputs -the machines and raw material that he buys, and the wages he's paying to his workers. So every capitalist constantly tries to get the most out of every dollar he spends, including from his workers. This is how firms survive in the market. It has nothing to do with greed.»*

Thanks!
Q&A



- **Engineering economics** is the application of economic principles to engineering decisions.
- Engineers constantly face choices between technically feasible alternatives: which machine to buy, whether to repair or replace equipment, which design to build.
- Engineering economics provides the tools to choose the alternative that is **economically best**, not just technically possible.
- *Engineering asks 'Can we build it?'*
  Engineering economics asks 'Should we build it, and which way is best?'